Digital Logic
Day 4

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uarch.space/slides/day4.pdf

Decimal Addition

$$
\begin{array}{r}
{\scriptstyle 1\ 1\ \ } \\
343 \\
+67 \\
\hline
410
\end{array}
$$

How do we add numbers?

Characteristics:

- Digits are added column by column (same place value).
- Results that exceed 9 are “carried” over.

Binary Addition

	1	1
1	1	110
+0	+1	+111
1	10	1101
1+0=1	1+1=2	6+7=13

Binary addition is similar to the addition we do.

You can also carry digits and add digits with the same place value together.

How could this be done with logic gates?

Binary Addition

Let’s look at just one column for now. Recall that 1 is ON and 0 is OFF.

$$\begin{array}{r} {}^{0}\;0 \\ +0 \\ \hline 0 \end{array} \qquad \begin{array}{r} {}^{0}\;0 \\ +1 \\ \hline 1 \end{array} \qquad \begin{array}{r} {}^{0}\;1 \\ +0 \\ \hline 1 \end{array} \qquad \begin{array}{r} {}^{1}\;1 \\ +1 \\ \hline 0 \end{array}$$

now you try

Half Adder

The Half Adder is a circuit that performs addition on two 1-bit numbers.

OUT 0: The output digit in the first column.

OUT 1: The carry digit.